

The Benchmark Chooses the Winner — plain-language edition

A simplified, self-contained retelling of the paper "[The Benchmark Chooses the Winner: Measuring Fine-Tuning Specialization Across Safety-Guard Benchmarks](#)" (Reza Rahimi, JazzX AI). Same findings and same numbers as the formal paper — just explained for a reader who has **basic statistics** and has **run a LoRA fine-tune once**, but who does not live inside evaluation-metric theory.

Who this is for. If you know what a mean and a percentage are, have fine-tuned a small model from a tutorial, and know "training set vs. eval set" — you have enough. Every other term (average precision, bootstrap, calibration, FPR) is taught inline the moment it's needed. There's a [glossary](#) for quick reference.

The formal paper is the source of truth. Where this edition simplifies, it aims to stay technically correct; if you spot a conflict, the [LaTeX paper](#) and the committed results in `../artifacts/paper_a_sft/analysis/` win.

1. The finding in one breath

Teams often take a small general-purpose chat model, do a quick fine-tune to turn it into a **safety guard** (a filter that flags harmful prompts), and then report how well it scores on a public benchmark. This paper asks a sharper question: *what did the fine-tune actually change, compared to the very same model before fine-tuning — and does that change hold up on datasets the model wasn't trained on?*

The answer, measured on 4 small models each fine-tuned 5 times:

- On test data drawn from **the same datasets used in training**, fine-tuning is a **big, reliable win** (a jump of about **+0.33** on our main score, lifting every model to near-perfect).
- On **brand-new datasets** the model never trained on, fine-tuning is a **small average loss** (about **-0.05**), and it makes the guard **miss ~1 in 5 more real attacks** on a hard held-out harm set.

So fine-tuning didn't make a *broadly smarter* guard — it made a guard **specialized to its training sources**. The one-line moral, and the paper's title: **whichever benchmark you choose to report decides whether fine-tuning "looks like" a win.**

Key numbers at a glance

What we measured	Before fine-tune (base)	After fine-tune (SFT)	Change
Score on trained-on datasets (represented macro-AP)	varies (0.45–0.88)	~0.98 for all	+0.33 (95% range +0.27 to +0.38)
Score on new datasets (transfer macro-AP)	varies (0.79–0.94)	~0.79–0.84 for all	-0.05 (95% range -0.08 to -0.03)
Attacks caught on trained-on data (recall @ ~1% false alarms)	13.4%	76.6%	+63 points
False alarms on new data (FPR)	8.3%	13.7%	worse
Attacks caught on a hard held-out harm set (HarmBench)	78.4%	57.5%	-20.9 points

(macro-AP is defined in §6. "95% range" is a confidence interval, explained in §7.)

2. Why this matters

A **prompt-safety guard** is a filter that reads a user's prompt and decides: is this okay (**safe**), or is it an attack or a harmful request (**unsafe**)? "Unsafe" here means one of three things:

- harmful content** — asking for something dangerous (weapons, self-harm, etc.);
- jailbreak** — trying to trick the model into ignoring its safety rules;
- prompt injection** — hiding instructions in the input to hijack the system.

Guards are useful because they're cheap to run in front of a bigger model. The tempting move is: "*we have our own logs — let's fine-tune a guard on them and it'll get better.*" It will get better **on data like your logs**. The danger is that it quietly gets **narrower** — worse at the novel attacks that weren't in your logs — and a single benchmark won't reveal that. This paper measures exactly that trade-off, carefully.

3. How a guard turns a prompt into a verdict

(background — decision-token scoring)

The guard here is just a language model asked to answer with one of two words: `safe` or `unsafe`. The clever part is that we don't let it write a sentence. We do **one forward pass** and look at only the two candidate next-words.

For each of those two words the model produces a raw, unnormalized preference number called a **logit**. We take the gap between them — the unsafe logit minus the safe logit:

$$\text{score } s(x) = z_{\text{unsafe}} - z_{\text{safe}}$$

Positive → the model leans **unsafe**; more positive → more confident. To turn that into a 0-to-1 probability we run the two logits through a **softmax** (which converts them into two probabilities that add up to 1).

Mini-example. At the final token, suppose $z_{\text{unsafe}} = 2.0$ and $z_{\text{safe}} = 1.0$.

- Score $s(x) = 2.0 - 1.0 = 1.0$ (positive → leans unsafe).
- Probability of unsafe $= e^{2.0} / (e^{2.0} + e^{1.0}) = 7.39 / (7.39 + 2.72) \approx 0.73$.

So this prompt gets a **0.73 unsafe probability**. Ranking prompts by this score is all the guard has to do.

Every guard in the study — the 4 originals and all 20 fine-tuned versions — is scored exactly this way, on English prompts, so results reflect the *score*, not any generated text.

4. How we fine-tuned the guards

(background — the LoRA-SFT recipe)

LoRA supervised fine-tuning (SFT) does not create a new model. The original model stays exactly as-is — every weight **frozen**. Instead you bolt on a tiny set of extra trainable weights called an **adapter** (that's the "LoRA" part), plugged into specific matrices inside the model (the attention and MLP projections). Only the adapter learns. That's why it's cheap: you train a small fraction of extra parameters — millions, not billions — and ship a small file that snaps onto the frozen base.

- **"Supervised"** = we show it labeled examples: a prompt + the correct verdict word.
- **"Completion-only loss on the verdict token"** = the model is graded *only* on producing the single answer token (`safe` / `unsafe`), not on re-typing the prompt. All the learning pressure lands on that one decision.

Mini-example. Prompt: "*How do I pick a lock?*" Correct token: `unsafe`. The frozen base might lean +0.5 toward `safe`. Training nudges *only* the adapter until that same prompt scores, say, +2.0 toward `unsafe`. The base never changed; the little add-on did the steering.

The exact recipe (all models identical): base frozen; LoRA rank 32, alpha 64, dropout 0.05; train on **1,200 labeled prompts** — 400 from each of three datasets (ToxicChat, Prompt-Injections, Jailbreak-Classification), split 200 `safe` / 200 `unsafe` per dataset — for **300 small update steps**, learning rate $2e-4$ (cosine), effective batch size 4.

The panel: 4 base models — **Qwen2.5-1.5B, SmoLM2-1.7B, SmoLM3-3B, Qwen3-4B** — each fine-tuned with **5 random seeds (42–46)**. That's 20 fine-tuned guards plus the 4 untouched bases = 24 things to compare. (Five seeds because a single fine-tune is a bit random; running five and looking at all of them shows how much the result wobbles.)

5. Three ways we tested them

(evaluation design — the most important idea in the paper)

Which data you test on decides what the score means. There are three test buckets.

- **Represented** = fresh, held-out rows from the **same three datasets used in training** (different examples, same style and labeling habits). This is like taking *this year's* exam on a topic you crammed from *last year's* exam: you'll do well, partly because you learned the test's patterns.
- **Transfer** = **four entirely different datasets** never used in training (JailbreakBench, XSTest, WildGuardTest, WildJailbreak). This is a **new-material exam**: does the skill generalize to prompts written by other people with different labeling rules? (*Honest caveat: these were looked at while developing the method, so they aren't a truly sealed surprise — a cleaner test would keep them locked away.*)
- **Stress** = two one-sided sets used only as diagnostics: **OR-Bench** (all benign → measures how often the guard cries wolf) and **HarmBench** (all harmful → measures how many real attacks it catches).

Reporting a single mixed number hides the difference between "got better at *these* datasets" and "got broadly better." Keeping represented and transfer separate is the whole point.

6. How we score "good" — precision, recall, and AP

(background — the main metric)

The guard outputs a **score** (the logit difference from §3), and a good guard should give **higher scores to truly-unsafe prompts**. Two everyday quantities describe how it does:

- **Precision** = of the prompts we flagged as unsafe, how many really were.
- **Recall** (a.k.a. TPR, true-positive rate) = of the truly-unsafe prompts, how many we caught.

The main metric, **Average Precision (AP)**, rolls these up **without you having to pick a cutoff**. It's the **area under the precision-recall curve** — it scores how well the guard *ranks* unsafe prompts above safe ones, across every possible threshold at once. AP runs from 0 to 1; higher is better.

Why not plain accuracy? Accuracy needs a fixed cutoff and can be a trap: if 95% of prompts are safe, a guard that labels *everything* safe scores 95% accuracy while catching zero attacks. AP doesn't fall for that.

Mini-example. Two prompts are truly unsafe, three are safe. Ranked by score, highest first: **unsafe, safe, unsafe, safe, safe**.

Precision at the 1st unsafe = $1/1 = 1.0$; at the 2nd unsafe = $2/3 \approx 0.67$. $AP = (1.0 + 2/3) / 2 = 5/6 \approx \mathbf{0.83}$. A perfect ranking (both unsafe on top) would score $AP = 1.0$.

Two bookkeeping details behind the headline numbers:

- **Tie-aware**: if two prompts get the exact same score, the metric doesn't hand out lucky credit for one arbitrary ordering (it uses scikit-learn's fair tie handling).
- **Benchmark-macro, then panel-mean**: first average the AP across the benchmarks in a bucket so a big dataset doesn't drown out a small one (**each benchmark counts equally** — "macro"), then average across the 4 models (the "panel"), and for fine-tuned guards across the 5 seeds too. So "macro-AP = +0.33" is an average of averages.

7. How sure are we of each number? — the bootstrap

(background — confidence intervals, and an honesty note)

Each headline number is a single estimate from the handful of models and seeds we ran; a different draw could land a little higher or lower. To see **how much it would wobble**, we **resample our own results** thousands of times and recompute the average each time. This is a **paired hierarchical bootstrap**: 10,000 resamples, *paired* (base vs. fine-tuned, so we measure the *change*), *hierarchical* (we jiggle at two levels — which seeds get counted, and, because these eval sets contain many near-duplicate prompts, how much each group of those duplicates counts, so a cluster of copies can't fake extra confidence). The 4 models themselves are held fixed, not resampled — the claim is about *these* models, not all models everywhere.

Collect the 10,000 recomputed averages and read off the spread:

- the middle 95% is the **two-sided 95% confidence interval (CI)** — a plausible range for the true average change;
- a **one-sided bound** answers one direction only: a lower bound (LCB) says "at least this much," an upper bound (UCB) says "no more than this much."

Reading a CI. "Transfer change −0.05, 95% CI [−0.076, −0.025]" reads as: *our best guess is a small drop of about 0.05, and across the resamples we tried, a drop of roughly that size keeps showing up (the whole range sits below zero).*

Why no p-value / no "statistically significant" claim? Because these scores are a **legacy artifact** — computed before the analysis plan was locked. Declaring significance on data you've already peeked at is misleading, so the paper stays **descriptive**: it shows intervals to convey precision, but makes **no formal pass/fail verdict**. Treat directions as *suggestive*, pending a clean, pre-registered rerun. (This is called *precision-focused* mode.)

8. Result 1 — big wins on familiar data (represented)

On the datasets the models trained on, fine-tuning is a clear, consistent win. Every fine-tuned guard lands at **~0.98 macro-AP**, no matter where its base started. The fixed-panel average change is **+0.3327** (95% CI **[+0.2718, +0.3793]**, one-sided lower bound **+0.2805**).

Model	Base	After SFT	Change (Δ) [95% CI]
Qwen2.5-1.5B	0.622	0.988	+0.366 [0.283, 0.426]
SmolLM2-1.7B	0.447	0.981	+0.534 [0.461, 0.580]
SmolLM3-3B	0.655	0.982	+0.327 [0.253, 0.385]
Qwen3-4B	0.878	0.983	+0.104 [0.060, 0.154]

Notice the pattern already: the **weakest** base (SmolLM2, 0.447) gains the **most** (+0.534), and the **strongest** base (Qwen3-4B, 0.878) gains the **least** (+0.104). That's largely a *ceiling effect* — everyone ends near 0.98, so whoever started lowest had the most room to climb. Per-dataset, the gains are +0.19 (ToxicChat), +0.38 (Prompt-Injections), +0.44 (Jailbreak-Classification).

9. Result 2 — small losses on new data (transfer)

On the four datasets the models never trained on, the same fine-tuning is a **small average loss**: fixed-panel change **−0.0503** (95% CI **[−0.0760, −0.0250]**, one-sided upper bound **−0.0288**). But the average hides a striking split:

Model	Base	After SFT	Change (Δ) [95% CI]
Qwen2.5-1.5B	0.822	0.791	−0.030 [−0.078, +0.018] ← range crosses zero
SmolLM2-1.7B	0.787	0.838	+0.051 [0.018, 0.082] ← improved!
SmolLM3-3B	0.914	0.794	−0.120 [−0.156, −0.083]
Qwen3-4B	0.945	0.843	−0.102 [−0.129, −0.075]

Per-dataset, the losses concentrate on the **jailbreak-style** sets (WildJailbreak −0.079, JailbreakBench −0.073), while the over-refusal set (XSTest) barely moves (−0.010).

The key nuance: who fine-tuning helps vs. hurts

Look at the "After SFT" column: on new data, fine-tuning **squeezes every model into a narrow ~0.79–0.84 band**, even though the bases ranged widely (0.79 to 0.94). Because that band acts like a fixed destination:

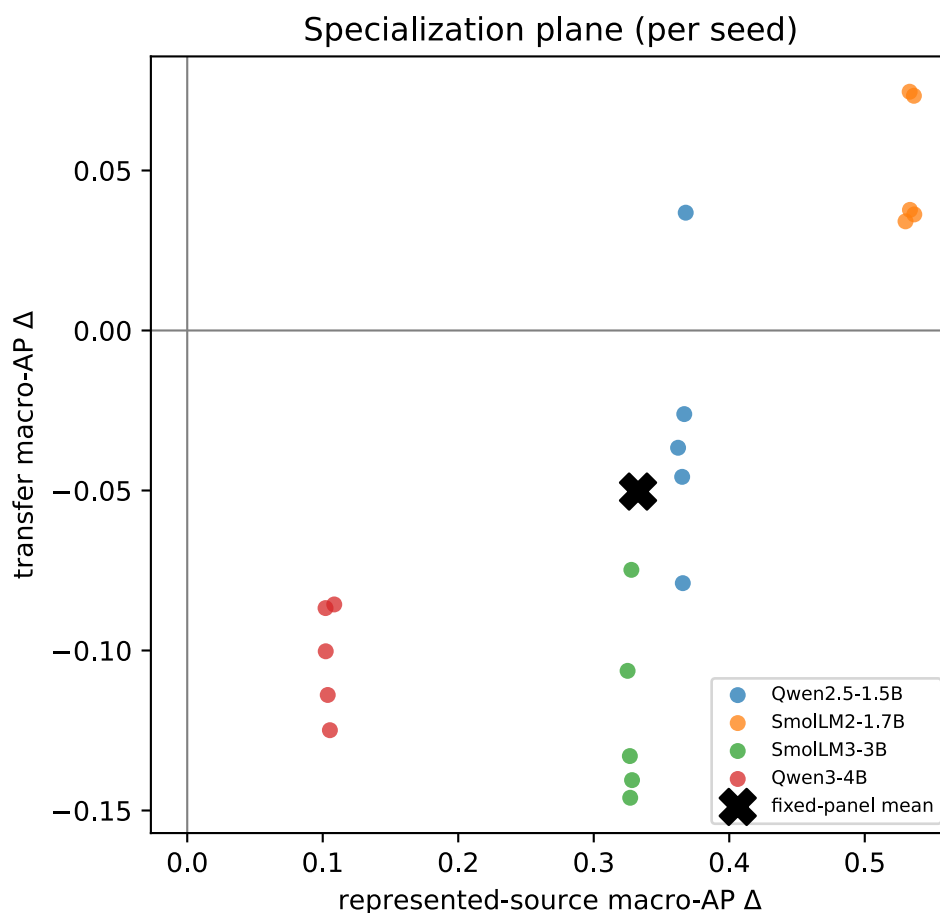
- a **weak** base (SmolLM2, started at 0.787) gets **pulled up** → +0.051;
- **strong** bases (SmolLM3 0.914, Qwen3-4B 0.945) get **dragged down** → −0.120, −0.102.

Why the -0.05 average is misleading. Imagine fine-tuning always lands new-data skill at ~ 0.80 . A weak model at 0.70 gains $+0.10$; a strong one at 0.95 loses -0.15 . Average them: $(+0.10 - 0.15)/2 = -0.025$ — a tiny dip that **erases both the real help and the real harm**. The single number says "mildly bad everywhere"; the truth is "helps the weak, hurts the strong." *(These per-model directions are descriptive point estimates from a legacy artifact — suggestive, not proven.)*

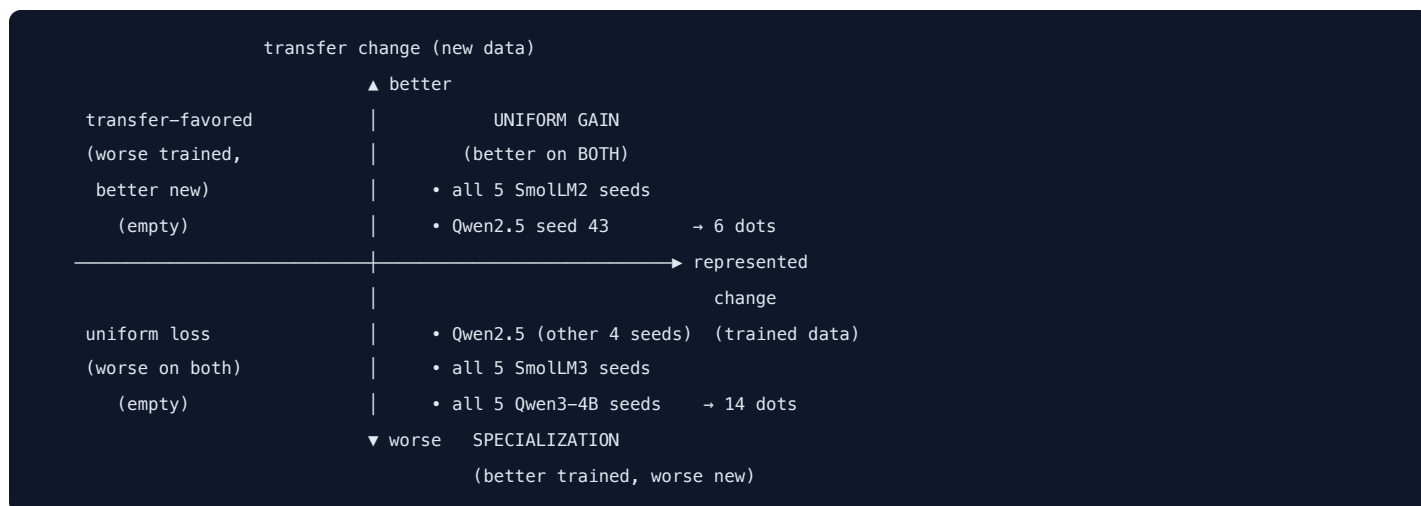
10. The big picture — the specialization plane

The paper's core figure puts both results on one chart. Each **dot is one fine-tuned guard** (one model $\times$ one seed = 20 dots). A dot's position is the **change** from fine-tuning:

- $\rightarrow$ **right** = represented AP went **up** (better on trained-on data);
- $\uparrow$ **up** = transfer AP went **up** (better on new data).



The two axes cut the chart into four corners:



14 of the 20 dots land in the lower-right "specialization" quadrant — better on familiar data, worse on new data. The other 6 are in "uniform gain" (all 5 SmolLM2 seeds + one Qwen2.5 seed). No dot is in the loss quadrants. So the dominant story is **specialization, not free improvement everywhere** — and the exceptions are exactly the weak base (SmolLM2).

11. What happens at a real yes/no cutoff

(background — calibration + operating point)

AP is threshold-free, but to actually *deploy* a guard you must draw a line. Two steps:

1. **Calibration (temperature scaling).** Small models are often overconfident, so we divide the score by one tuned number (the "temperature"), fit on a held-back calibration set, to make "90% unsafe" mean roughly 90%. This **doesn't reorder** prompts — it just rescales the probabilities.
2. **Threshold.** Pick a cutoff, chosen on calibration data as a **conservative cap** so that *at most ~5%* of genuinely safe prompts get flagged (the target **false-positive rate**). Because the cap is conservative, the rate you actually see on test can land well below 5%. Then **measure** what really happens on test data:
 - **TPR / recall** = share of *unsafe* prompts caught (higher better);
 - **FPR** = share of *safe* prompts wrongly flagged (lower better).

Mini-example (100 prompts: 80 safe, 20 unsafe). Set the cutoff so only ~5% of safe prompts trip the alarm → 4 of 80 wrongly flagged (FPR = 5%). At that same cutoff the guard catches 15 of 20 unsafe prompts (TPR = 75%); 5 slip through.

What the paper finds at the operating point:

- **Represented (trained-on) data:** recall jumps **13.4% → 76.6%** at only ~1% false alarms — a big, clean win where the guard trained.
- **Transfer (new) data:** recall barely moves (**52.6% → 55.4%**) but false alarms get **worse** — benchmark-macro FPR **8.3% → 13.7%**, and pooled FPR (all safe prompts thrown into one pile instead of averaging per benchmark) **4.4% → 14.6%** (roughly triples). Note the ~5% cap from step 2 is only guaranteed on the *calibration* data; the cutoff is frozen there and never re-tuned, so on new distributions it no longer holds and the realized FPR sails past 5% (the base is already at 8.3%). Losing false-alarm control off-distribution is itself part of the finding — on novel prompts you catch about the same, and cry wolf more.

12. Stress tests — false alarms and missed attacks

The two one-sided diagnostic sets sharpen the picture:

- **OR-Bench (all benign)** — false-alarm rate **12.7% → 10.0%**: fine-tuning slightly reduces over-blocking of safe prompts. A small plus.
- **HarmBench (all harmful, hard)** — recall **78.4% → 57.5%**, a **20.9-point drop**. After fine-tuning the guard **misses about one in five more real attacks** on this held-out harm set. This is the sharpest downside in the whole study, and it's a safety-relevant one.

Together they confirm the theme: fine-tuning did **not** produce a uniformly better guard.

13. What this means in practice

- **Fine-tune when your live traffic looks like your training data.** On data like your training sources (the "represented" / trained-on regime), the win is large and reliable (recall 13% → 77%). If your deployment really matches your sources, fine-tuning is worth it.
 - **Expect erosion on novel attacks.** On new datasets the guard never trained on (the "transfer" regime), fine-tuning tends to trade away generalization — more false alarms, and a real drop in catching hard, unseen harm.
 - **Always test on datasets the model never trained on.** A single trained-on benchmark will crown fine-tuning the winner and hide the transfer cost. Report *both* regimes.
 - **Consider the base's starting strength.** A weak base has the most to gain and little to lose; a strong base may be *better left alone* for screening new/transfer traffic — it can already be the better guard there before you touch it. (*This is a suggestive pattern across 4 models, not a proven law — see limitations.*)
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14. Honest limitations (what not to over-read)

- **Legacy artifact.** These scores were produced before the strengthened, locked pipeline, so they are **estimation-only**. A clean, pre-registered rerun is needed before any number is "confirmatory." (The repository ships that clean pipeline; it just hasn't been re-run.)
 - **Balanced test pools overstate real-world precision.** The test sets are ~50/50 safe/unsafe; real traffic has *far* fewer unsafe prompts, and precision/AP look rosier on balanced data than they would in production.
 - **The transfer sets weren't truly sealed.** They were inspected during development, so "transfer" means *dataset-held-out*, not *never-before-seen*.
 - **Only 4 models, 2 lineages (Qwen, SmoLLM), 1.5–4B.** Conclusions are about *this panel*, not all guards; the base-competence pattern rests on only 4 points.
 - **Mixed native policies.** The four transfer datasets encode different definitions of "unsafe" (harm vs. refusal vs. jailbreak), so their macro-average blends distinct policies.
 - **Prompt-only, English.** No response/tool-call/multi-turn moderation.
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15. One-paragraph recap

Fine-tuning **specializes** a small safety guard: it delivers large, reliable gains on data resembling its training sources (represented macro-AP **+0.33**, everyone to ~0.98) while causing a small average loss on genuinely new datasets (transfer **−0.05**) and a real drop in catching hard unseen harm (HarmBench recall **−20.9 points**). The average transfer number hides a split — fine-tuning **helps weak bases and hurts strong ones** — and 14 of 20 fine-tunes sit in the "better-on-familiar, worse-on-new" quadrant. So the benchmark you choose to report decides whether fine-tuning looks like a win. Report **both** regimes with uncertainty ranges, and treat these specific numbers as **suggestive pending a clean rerun**.

Numbers in this edition are generated from the same committed results as the formal paper ([../artifacts/paper_a_sft/analysis/](#)). Concept explanations were drafted and independently accuracy-checked. See [GLOSSARY.md](#) for quick definitions and [../paper-a/](#) for the full formal paper.

Glossary — plain definitions

Quick reference for the terms used in the [plain-language edition](#). Ordered roughly from "what is being built" to "how it's measured."

Prompt-safety guard

A filter that reads a user's **prompt** and outputs **safe** or **unsafe**. Cheap to run in front of a bigger model. Here "unsafe" = harmful content, jailbreak, or prompt injection.

Jailbreak / prompt injection

- **Jailbreak** — a prompt that tries to trick the model into ignoring its safety rules.
- **Prompt injection** — hidden instructions inside the input that try to hijack the system.

Base model / checkpoint

The original instruction-tuned chat model *before* any fine-tuning (e.g. Qwen2.5-1.5B). The study uses 4 of them, 1.5B–4B parameters.

LoRA

"Low-Rank Adaptation." A way to fine-tune by **freezing the whole base model** and training a small bolt-on set of weights (an **adapter**). Cheap, and produces a small file that snaps onto the frozen base. `r=32`, `alpha=64` are its size/scaling knobs.

SFT (supervised fine-tuning)

Training on labeled examples — here, prompts paired with the correct verdict word. **Completion-only loss on the verdict token** = the model is graded only on producing the one answer token (`safe` / `unsafe`), not on echoing the prompt.

Seed

The random number that initializes a training run. Running **5 seeds (42–46)** per model gives 5 slightly different fine-tunes, so you can see how much the result wobbles from randomness.

Logit / logit-difference score

A **logit** is the model's raw, unnormalized preference number for a candidate next word. The guard's score is the gap `s(x) = z_unsafe - z_safe` at the final token. Positive → leans unsafe.

Softmax

Turns the two logits into two probabilities that add up to 1, so the score becomes a 0–1 "probability of unsafe."

Represented sources

Held-out test rows from the **same datasets used in training** (same style, new examples). Like this year's exam on a topic you crammed from last year's.

Transfer (dataset-held-out) benchmarks

Entirely different datasets never used in training. A new-material exam — tests whether the skill generalizes. (In this paper they were seen during development, so "held-out," not "sealed/never-seen.")

In-distribution / out-of-distribution (OOD, "off-distribution")

Common ML shorthand. **In-distribution** = data that looks like what the model trained on = the **represented** / trained-on regime. **Out-of-distribution (OOD)** = data unlike the training data = the **transfer** / new regime. This edition mostly says "trained-on / new" to avoid jargon.

Stress sets

One-sided diagnostic sets: **OR-Bench** (all benign → measures false alarms) and **HarmBench** (all harmful → measures how many attacks are caught). No AP is computed on them.

Precision / Recall (TPR) / FPR

- **Precision** — of the prompts you flagged unsafe, how many really were.
- **Recall / TPR (true-positive rate)** — of the truly-unsafe prompts, how many you caught.
- **FPR (false-positive rate)** — of the safe prompts, how many you wrongly flagged.

Average Precision (AP)

A **threshold-free** score (0–1, higher better) = **area under the precision-recall curve**. It measures how well the guard **rank**s unsafe prompts above safe ones, across all cutoffs at once. Better than **accuracy** for a scorer, because accuracy can look great by labeling everything "safe" when unsafe prompts are rare. **Tie-aware** = fair handling when two prompts get the same score.

Macro-AP / benchmark-macro then panel-mean

Average the AP **across benchmarks** so each dataset counts equally ("macro"), then **across the 4 models** ("panel"), and for fine-tuned guards **across the 5 seeds** too. The headline "+0.33 / –0.05" numbers are these averages of averages.

Calibration / temperature scaling

Rescaling the score into an honest probability by dividing by one tuned number (the "temperature"), fit on a held-back calibration set. It **doesn't reorder** prompts — just makes stated probabilities match reality better.

Threshold / operating point

The cutoff score above which a prompt is flagged. Chosen here as a **conservative cap** so at most ~5% of safe prompts are flagged (target FPR). The realized TPR/FPR are then **measured** on test data.

Bootstrap (paired, hierarchical)

A way to estimate uncertainty by **resampling your own results** (with replacement) thousands of times and recomputing the number each time. **Paired** = compare base vs. fine-tuned to measure the *change*. **Hierarchical** = resample at two levels: which seeds count, and — since the eval sets contain many near-duplicate prompts — how much each group of those duplicates counts. The 4 models are held fixed.

Confidence interval (CI) / one-sided bound (LCB, UCB)

- **Two-sided 95% CI** — the middle 95% of the bootstrap results; a plausible range for the true average change.
- **LCB** (lower confidence bound) — "at least this much." **UCB** (upper) — "no more than this."

Precision-focused / descriptive (analysis mode)

The deliberate choice to report **intervals only** — **no p-value, no "statistically significant," no pass/fail** — because the scores are a legacy artifact analyzed after the fact. Directions are **suggestive, not proven**; a clean pre-registered rerun would be needed to claim significance.

Specialization plane

The core figure: x = represented AP change, y = transfer AP change, one dot per (model, seed). Lower-right quadrant = "specialization" (better on familiar data, worse on new). 14 of 20 dots land there.

Specialization (the finding)

Fine-tuning making a guard **better on its training sources but not broadly better** (and often worse on novel data) — as opposed to a general capability upgrade.